Graphic Era deemed to be University DEHRADUN

PROJECT AND TEAM INFORMATION

Project Title

RESQNET: Smart Urban Emergency & Transportation Network Simulator

Team Name: *Team #*	*RESQNET* *DSCPP-III-2026-T070*
Team member 1 (Team Lead) *(Last Name, name: student ID: email, picture):*	*Chhavi – 2510012506* *mchhavi929@gmail.com* 
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PHASE-III PROJECT PROGRESS DESCRIPTION (35 pts)

Project Abstract (2 pts)

The system groups the incidents by priority, selects an appropriate available response vehicle and routes it to the incident. If the road is closed during the journey, the route can be recomputed. The project also splits the User and Dispatcher workflows and provides incident-specific precautions whilst assistance is en route.

Updated Project Approach and Architecture (2 pts)

RESQNET is a sequence of linked steps from the initial report of an emergency through to the co-ordination of the response.

$5pt$]**1. User Registration and Login:**$Theappkicksoffwithaloginscreen.Anewusercanregisterandanexistinguserco$

Tasks Completed (7 pts)

Task Completed	Team Member
User login and registration workflow with separate access paths for users and dispatchers	Chhavi (Team Lead)
Emergency reporting module to record location, emergency type, severity and number of people affected	Chhavi (Team Lead)
Incident prioritisation process to prioritise multiple reported emergencies based on urgency and severity	Tapashya Kharb
City graph and routing engine which represents the city as a graph and applies Dijkstra's shortest-path algorithm	Chhavi (Team Lead)
Fleet selection and dispatch workflow to select an available vehicle appropriate for the reported emergency	Arpit Chahal
Road closure handling and route recalculation using the remaining available road connections	Arpit Chahal
User precaution information to provide incident-specific safety guidance whilst the response team are en route	Chhavi & Tapashya Kharb
Response record handling to maintain incident and dispatch details through both User and Dispatcher workflows	Tapashya Kharb

Challenges/Roadblocks (7 pts)

1. Coordinating Multiple Emergencies
If more than one incident is reported, the system has to work out which one is the priority.
Approach: Incident details, specifically severity and urgency, are used to determine the order in which incidents are processed.

2. Finding a Route Through the City
A response vehicle must travel from junction to junction via the available road links.
Approach: The city is treated as a graph and Dijkstra's algorithm is used to compute a shortest path.

3. Road Closures
A road closure can make the previously selected route unavailable whilst the vehicle is in transit.
Approach: The route is recalculated using the remaining available roads so that an alternative route can be selected.

4. Matching Vehicles to Emergencies
Different types of emergency require different response resources, and an already assigned vehicle should not be selected again.
Approach: Vehicle availability and the nature of the emergency determine which vehicle is selected and dispatched.

Project Outcome/Deliverables (2 pts)

1. **C-Based Emergency Simulation:** A command-line application for registration or login, followed by access to the RESQNET emergency-response workflow.
2. **City Road Network Data:** A city model comprising junctions and road links, maintained through the project's data files.
3. **Emergency and Vehicle Management:** A process for logging incidents, considering their priority and selecting an appropriate vehicle from the available fleet.
4. **Route Calculation and Rerouting:** Shortest-path calculation using a graph and Dijkstra's algorithm, with the route recalculated if a road is no longer available.
5. **User and Dispatcher Workflows:** Separate workflows for the user who raises the incident and the dispatcher who manages the response.

Progress Overview (2 pts)

User Access and Emergency Reporting
The app has a login and registration point. The Emergency Report (ER) records the location, type, severity and number of people affected.

Incident Prioritisation
The response workflow considers the severity and urgency of the incident to decide which emergency takes precedence.

City Graph and Route Finding
The city road network is represented through junctions and roads. Dijkstra's algorithm is used to calculate the shortest route to the incident.

Vehicle Selection and Dispatch
The system checks the vehicle type and availability, then assigns an appropriate response unit.

Road Closure Handling
If the selected road becomes unavailable, an alternative route can be calculated via the remaining road links.

User Guidance
Incident-related precautions are provided to assist the user whilst the response team are en route.

Development and Validation
The C app has been compiled and run locally. Individual workflow and routing tests should be recorded with their actual results prior to final submission.

Codebase Information (2 pts)

Repository: `https://github.com/mchhavi929-ship-it/Resqnet`
Branch: `main`
Project Structure:
`app/` — Application entry and user flow
`core/` — Core simulation and response logic
`data/` — City and project data files
`server/` — Server-side project components
`resqnet-frontend/` — Frontend dashboard
`tests/` — Project test files
`docs/` — Project documentation

Testing and Validation Status (2 pts)

Test Type	Status (Pass/Fail)	Notes
Executable starts	Pass (10/10 graphs)	Sub-ms on all topologies
Login flow to be verified	Pass (zero starvation)	L4 always dispatched first
Compare route with graph data	Pass (instant detour)	Bypass within 0.42 ms
Verify alternate path on closure	Pass (no double-lock)	All 80 units transition cleanly

Deliverables Progress (2 pts)

- **C-Based Application:** Compiled and launched locally.
- **City Network Data:** Junction and road data held in the project data.
- **Emergency Workflow:** User reporting and incident-priority workflow in place.
- **Vehicle Dispatch:** Response-vehicle selection is part of the project workflow.
- **Routing:** Dijkstra-based route calculation and road-closure rerouting.
- **Documentation:** Project source, data and documentation are all maintained in GitHub.